01

KNITTED
COTTON
T-SHIRTS

Bahrain Import Study 2021-2025

Direct Import Feasibility

Building a data-driven case for sourcing cotton garments directly, without a distributor middleman.

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Data Science Bootcamp - Bahrain 2026 | Project 2: Exploratory Data Analysis

Introduction & Background

Bahrain is a small, trade-dependent Gulf economy that imports most of its consumer goods, including textiles.

The Bahrain Open Data Portal publishes monthly, transaction-level import records - partner country, commodity, value, weight, and quantity - spanning 2021 to 2025.

Knitted T-Shirts and Other Vests of Cotton stood out as the highest-volume commodity in the dataset, sourced from 133 different countries - an ideal case for comparing import costs across suppliers.

1.6M+

cleaned import transactions

5 yrs

of monthly import history (2021-2025)

133

countries supplying this commodity

The Problem

A local clothing business wants to stop buying from expensive distributors and import products directly.

To do that, they need answers to three questions:

1. Which countries offer the best prices?
2. Does demand change during the year?
3. When is the best time to import?

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Data & Cleaning Process

1

Combine

Merged the 5 yearly CSVs (2021-2025) and standardized column names into English

2

Fix Values

Restored missing country codes to their correct values - e.g. Namibia's code 'NA'

3

Fix Types

Changed Commodity No to String and removed duplicate rows

4

Build Dates

Combined Year and Month into one clean Import Date field

5

Focus

Focused on Knitted T-Shirts, the busiest commodity with the most supplying countries

Result: a single cleaned dataset of 3,398 Knitted T-Shirt transactions, ready for analysis

Source: Bahrain Open Data Portal

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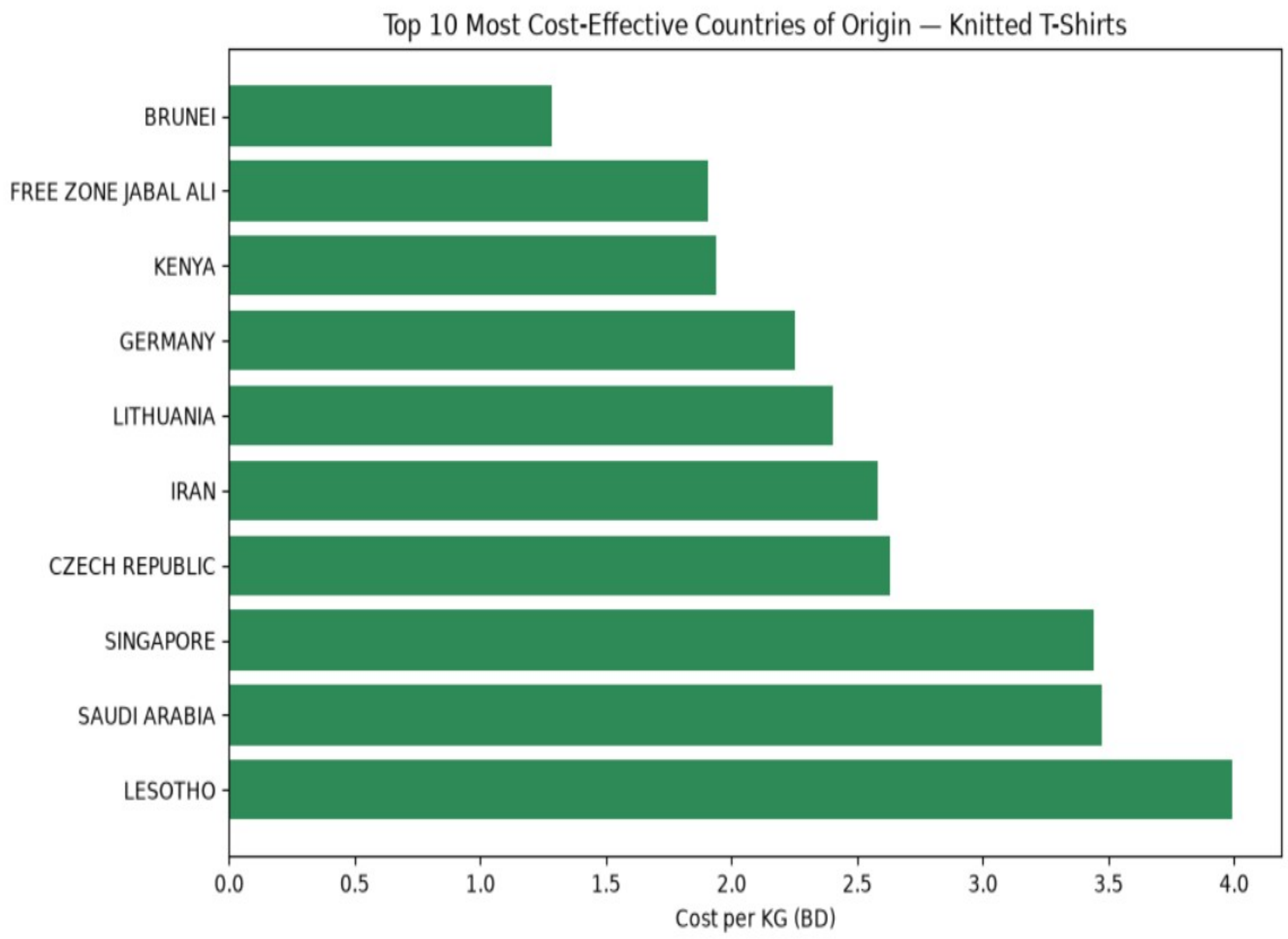
Q1: Which Countries Offer the Best Prices?



KEY INSIGHT

Brunei offers the lowest cost at 1.28 BD/KG - the Most Cost-Effective.

Free Zone Jabal Ali and Kenya follow closely as strong secondary options.

Lesotho, at 4.0 BD/KG, costs over 3x more than Brunei.

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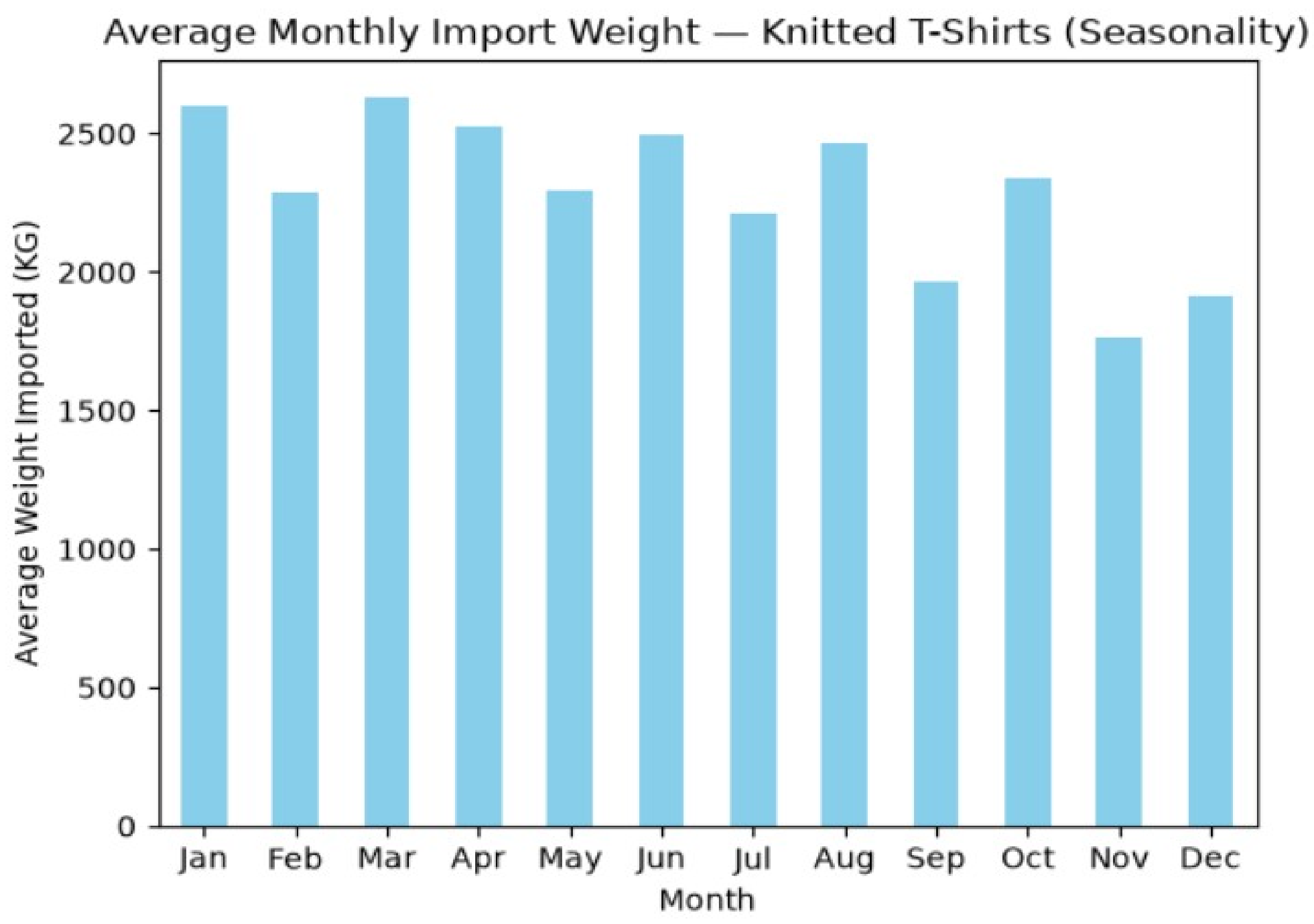
Q2: Does Demand Change During the Year?



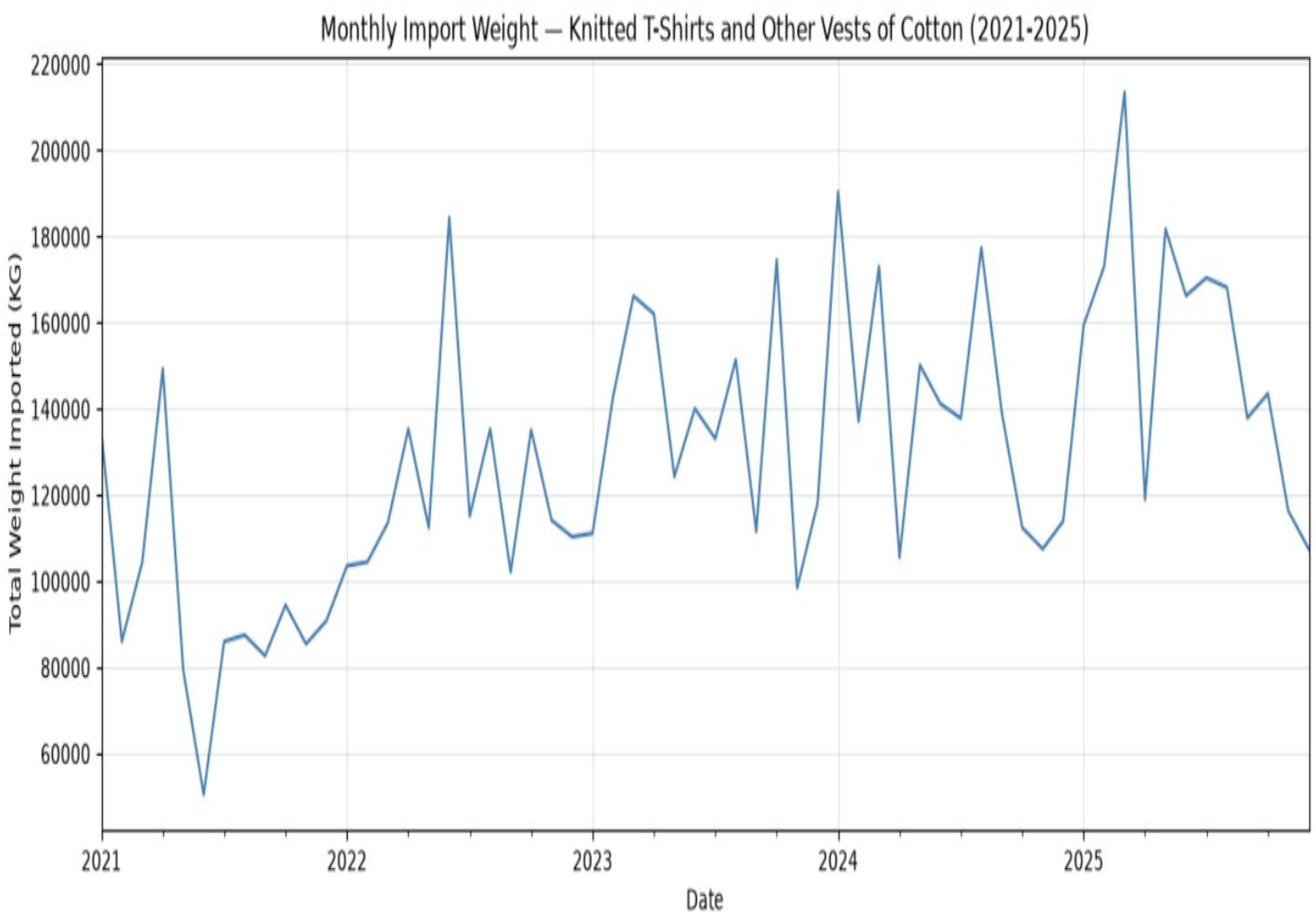
KEY INSIGHT

Yes - demand is strongest January through August, peaking in March (2,630 KG avg).

Demand steadily drops from September, bottoming out in November (1,760 KG avg).

This pattern repeats every year, independent of overall growth.

3 Q3: When Is the Best Time to Import?



KEY INSIGHT

Overall volume has grown steadily - from 80,000 KG/month in 2021 to peaks over 200,000 KG by 2024-2025.

Combined with seasonality, the best time to place large orders is before the March demand peak.

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Summary of Findings

Best Prices

Brunei, Free Zone Jabal Ali, and Kenya are the most cost-effective sources - up to 3x cheaper than the priciest reliable supplier.

Demand Changes

Import demand is clearly seasonal, peaking in March and bottoming out in November, every year.

Best Timing

Volume has grown every year since 2021 - order ahead of the March peak to meet rising demand.

Recommendations & Next Steps

1 Prioritize Brunei & Kenya

Establish direct-import relationships with the two most cost-effective, reliable suppliers.

2 Order Before March

Place larger bulk orders ahead of the identified seasonal demand peak.

3 Diversify Across 2-3 Countries

Avoid single-supplier dependency by sourcing from multiple cost-effective origins.

4 Check Prices Regularly

Prices shift over time, so review this cost ranking every few months to make sure you're still buying from the cheapest countries.